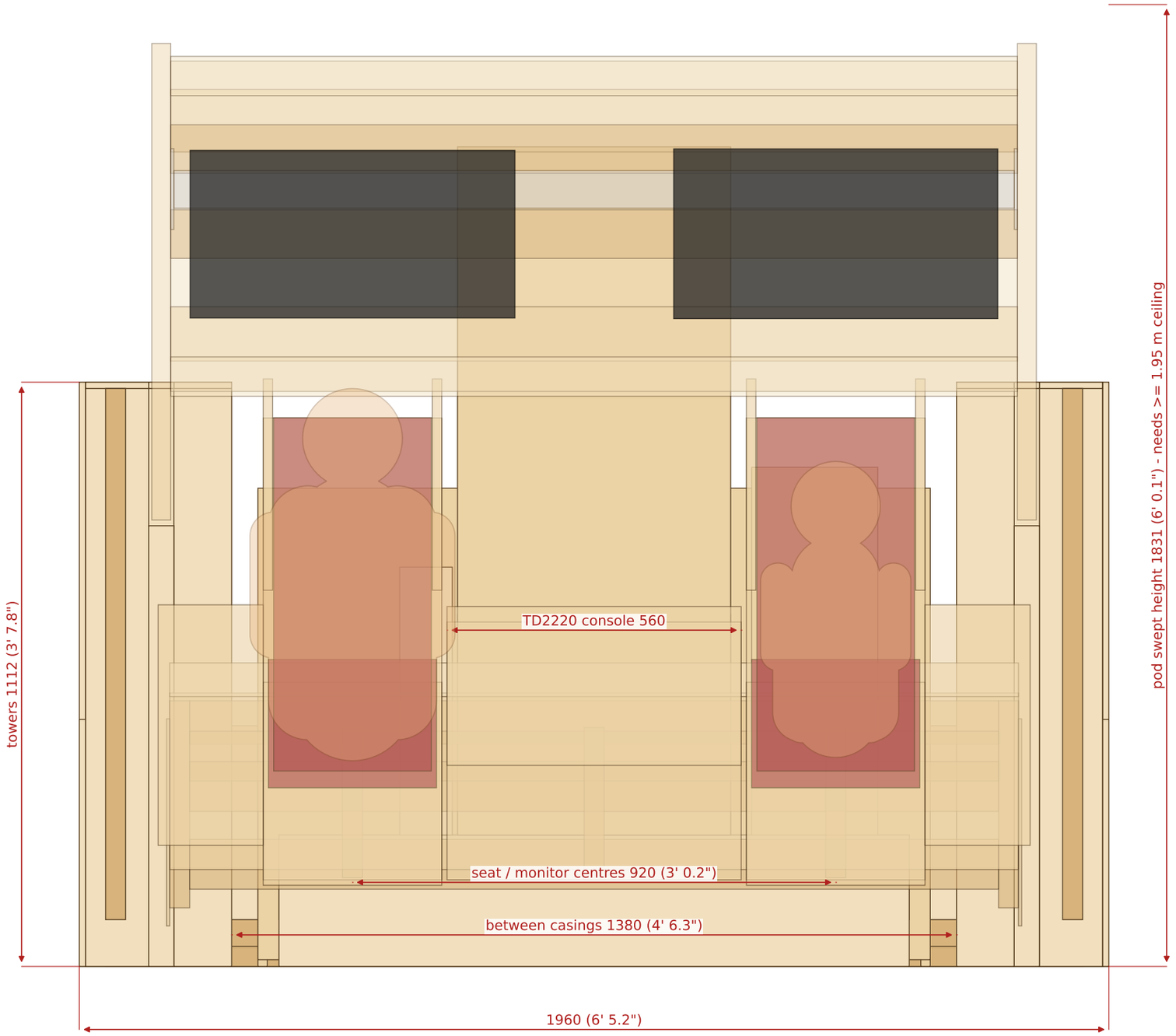
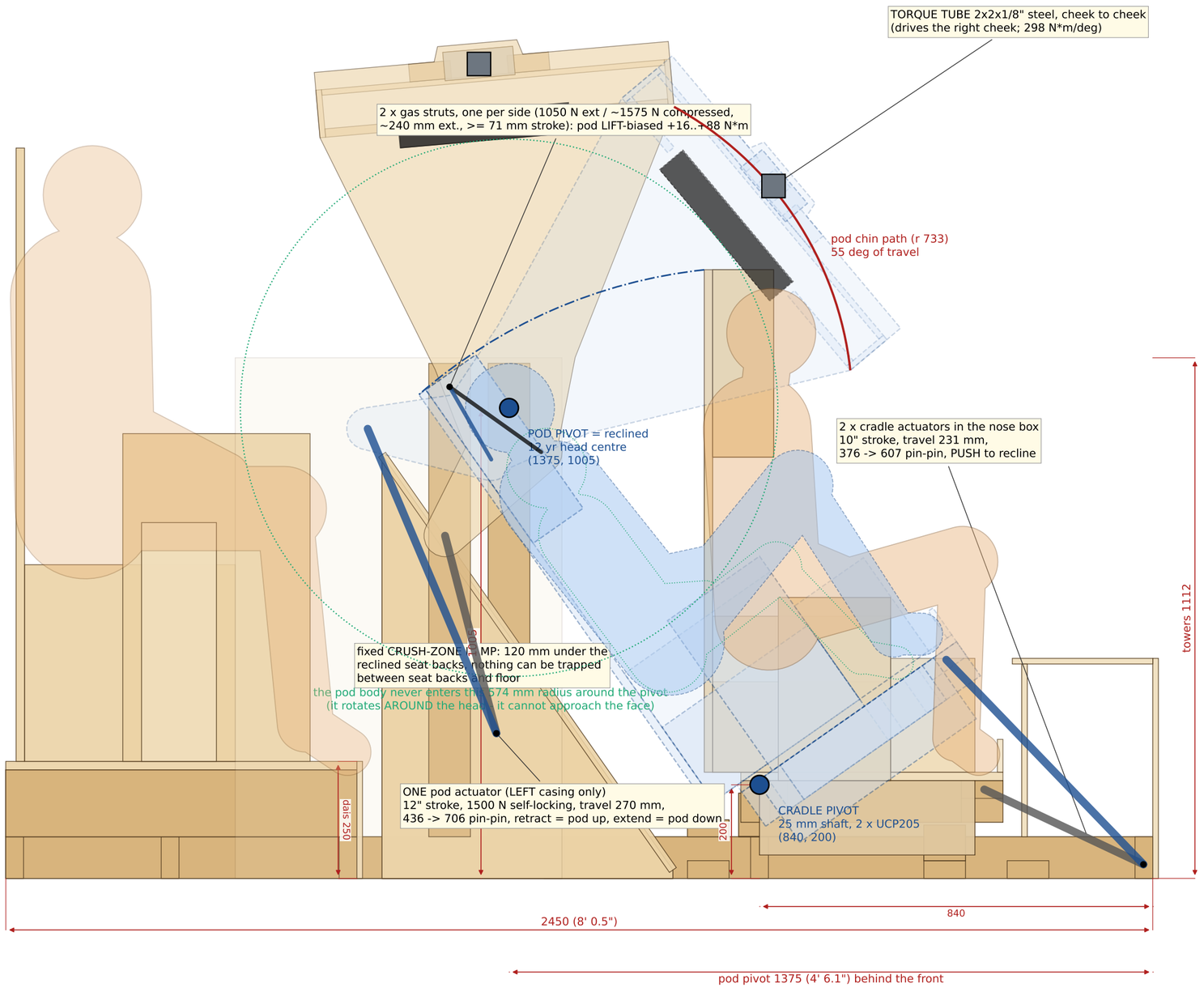
1. Front elevation, FLIGHT position (cradle reclined 35 deg, pod down) - nose box front skin removed

Pod (blue-grey) carries both monitors, the 4 dash panels (lower band) and the 3 overhead panels (top band). The cradle carries the seats, the TD2220 centre console, both side consoles and the sticks. Dad's dais (behind) is fixed.

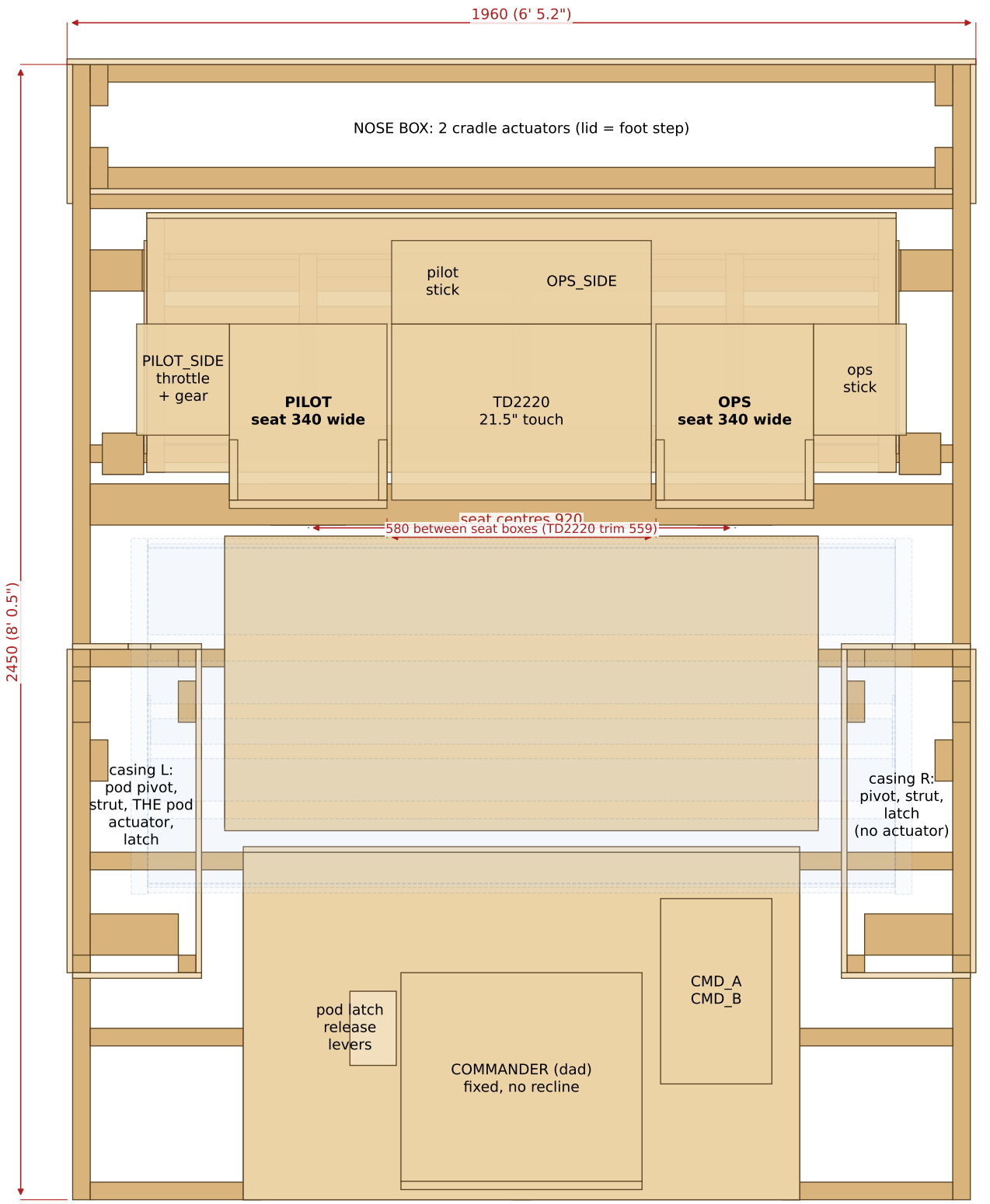


2. Side elevation + MOTION GEOMETRY (from the right, front at the right). Solid = BOARDING (upright, pod up 95 deg), blue dashed = FLIGHT (recline 35 deg, pod down 40 deg)

MOTION: hold FLIGHT -> cradle reclines 0 -> 35 deg (~39 s), then the pod swings down 95 -> 40 deg (~34 s). Hold BOARD -> pod up first, then seats upright.
Clearances at the hard stops (12 yr manikin = worst case): pod chin to thighs/shins 195 mm (ISO 13854 leg >= 180), pod to head 251 mm, pod parked UP: lowest point over the seats z = 1639 mm.
Eye -> screen centre: 12 yr 542 mm, 6 yr 590 mm. Pod mass 52 kg (model), CG 613 mm from its pivot.



3. Plan view, BOARDING position (from above, front at the top; pod parked up shown dashed)



4. Mechanism sizing, locks, hard stops and the hardwired interlock

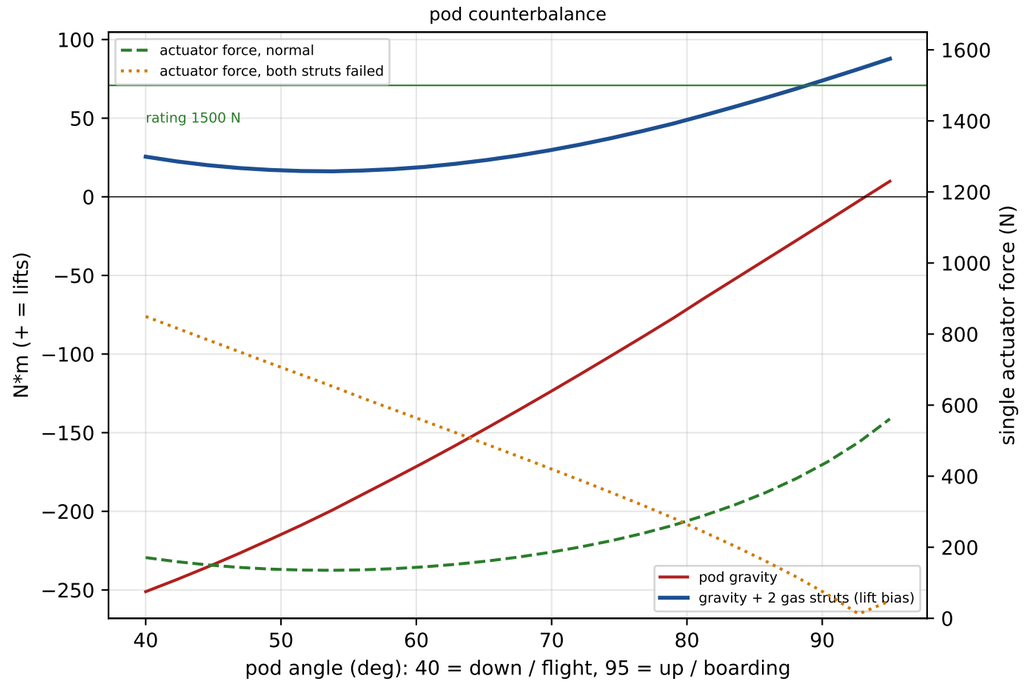
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CRADLE (both kid seats, consoles, TD2220) - recline 0 -> 35 deg about the 25 mm shaft at (840, 200)
  design load: 2 occupants x 50 kg + cradle ~158 kg  -> gravity torque -263 .. +270 N*m
  (negative = wants to sit up: rests on the front rubber pads; positive = wants to lie back: rests on the recline stop blocks)
  2 actuators, 10" (254 mm) stroke, retracted ~364 pin-pin; used travel 376 -> 607 mm
  peak force 775 N EACH (incl. +40 N*m friction) -> BUY 2 x 1500 N (330 lbf) 12 V, self-locking (acme), ~5-8 mm/s

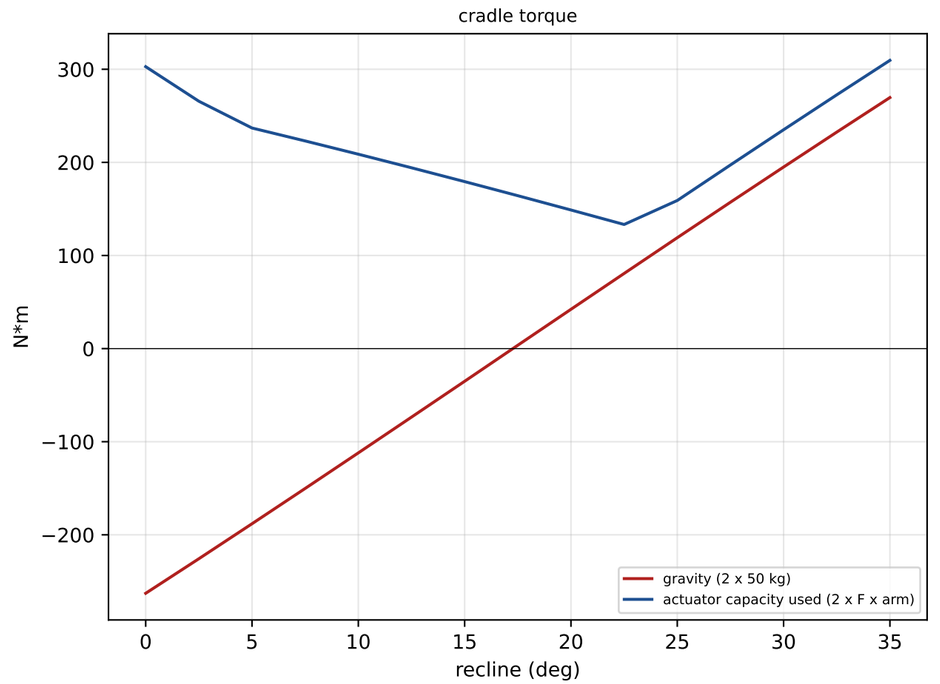
POD (2 monitors, 7 panels, torque tube, ~52 kg) - swings 95 -> 40 deg about (1375, 1005) = the reclined 12 yr head centre
  gravity torque -251 .. +10 N*m;  with the 2 gas struts: +16 .. +88 N*m
  -> LIFT-BIASED over the whole travel: if the actuator or its clevis pin fails the pod rises to the UP stop (away from the kids);
     at DOWN the spring pawls + the self-locking actuator hold it; at UP the struts press it on the stop and the pawls lock it.
  gas struts: 2 x ~1050 N (F1, extended, progression 1.5), ~240 mm extended, >= 71 mm stroke, M8 eye ends, rod DOWN, damped
  ONE pod actuator (left casing, x = -773), 12" stroke (retracted ~420), travel 436 -> 706 mm, pin (1677, 960), anchor (1402, 310)
  force: normal (2 struts, +25 N*m friction) <= 561 N;  one strut failed 466 N;  BOTH struts failed 850 N
  -> BUY 1 x 1500 N (337 lbf) 12 V self-locking, <= 10 mm/s (~27 s); margin 1.8x struts-failed, 2.7x normal
  TORQUE TUBE 2 x 2 x 1/8 in square steel tube (50.8 x 50.8 x 3.2), 1600 mm cheek to cheek, J = 345e3 mm4, 298 N*m/deg:
    normal (2 struts)                          44 N*m, twist 0.15 deg, chin  2.1 mm, 3 MPa
    far-side strut failed                     126 N*m, twist 0.42 deg, chin  6.0 mm, 9 MPa
    both struts failed                        126 N*m, twist 0.42 deg, chin  6.0 mm, 9 MPa
    actuator stall at rating vs far-side stop 458 N*m, twist 1.53 deg, chin 22.0 mm, 32 MPa
    ends: welded 6 mm plates, 4 x M10 through each 36 mm cheek + 6 mm backing plate.
    Set the RIGHT cheek's stops 1-2 mm 'later' than the left so the actuator never twists the tube
    against a stop; the far side is carried by the tube, not by racking the cheeks.

HARD STOPS (steel, bolted; they - not the actuators - define the end positions):
  cradle upright: rails on 6 mm rubber pads (z 114);  cradle reclined: seat boxes land on rubber stop blocks at 35 deg (+0.5)
  pod DOWN: 6 mm steel stop pin on each cheek crank lands on a rubber-faced steel angle in the casing at 40 deg
  pod UP: same pin on a second stop at 95 deg.  External limit switches stop the actuators 2-3 mm before each stop.

LOCKS (mechanical, both ends):
  pod: spring-loaded steel pawl in each casing drops into a notch in a 6 mm steel latch plate on the cheek at UP and at DOWN
       (ramped like a door latch: it clicks in by itself).  Dad releases both pawls with one Bowden-cable lever LEFT of his seat;
       a micro switch on the lever is in series with the pod motors -> the pod can only move while dad holds the lever
       AND the switch (two-hand hold-to-run).  'Pawl engaged' micro switches feed the interlock below.
  cradle: self-locking actuators + stops; the pod interlock keeps it from moving under a lowered pod.

HARDWIRED INTERLOCK (no microcontroller can move anything; the game only READS 3 isolated senses):
  12V+ --[20A fuse]--[MOTION key]--[E-STOP NC]--+--CREW POS (ON)-OFF-(ON) hold-to-run DPDT (drives 4 relays)
  ONE pod limit set: POD_UP + POD_DOWN roller switches on the LEFT (driven) cheek only; the right side has only its pawl switch.
  FLIGHT (toggle up):
    cradle RECLINE runs  while  POD_UP_LOCKED (both pawls)  AND  NOT CRADLE_RECLINED limit
    pod DOWN runs        while  CRADLE_RECLINED (both sides) AND  release lever held  AND  NOT POD_DOWN limit
  BOARD (toggle down):
    pod UP runs          while  release lever held  AND  NOT POD_UP limit
    cradle UPRIGHT runs  while  POD_UP limit AND POD_UP_LOCKED  AND  NOT CRADLE_UPRIGHT limit
=> the pod can never come down unless the seats are fully reclined, and the seats never move under a lowered pod.
Pod chin carries a pressure-sensitive SAFETY EDGE (NC) in series with 'pod DOWN'.
Limit switches: roller-lever micro switches (NC) on the stops, 1N5408 diode across each so you can always reverse.
Relays: 4 x 30 A automotive SPDT (Bosch-type) wired as 2 H-bridges (cradle: 2 actuators in parallel; pod: 1 actuator); 12 V 20 A supply.
Position senses (READ-ONLY, 3 separate NO micro switches, PC817 optos -> Falcon Command GPIO39/40/41):
  CRADLE_RECLINED, POD_DOWN_LOCKED (pawl in DOWN notch), POD_UP_LOCKED (optional). Never in the motor path.
```





5. Pinch points, guards and computed clearances

Computed by frame_model.py from the CAD (manikins: 95th pct 12 yr / 50th pct 6 yr, capsule bodies).

POD (sweeping 95 -> 40 deg) vs the RECLINED kids - closest approach anywhere in the travel:

PILOT upper arm	155 mm	(pod at 95.0 deg,	Pod cheek L)
PILOT thigh	195 mm	(pod at 40.0 deg,	Pod chin plate)
PILOT shin	220 mm	(pod at 40.0 deg,	Pod chin plate)
PILOT torso	233 mm	(pod at 95.0 deg,	Pod cheek L)
OPS upper arm	241 mm	(pod at 95.0 deg,	Pod cheek R)
PILOT head	251 mm	(pod at 40.0 deg,	Pod cheek L)
OPS head	261 mm	(pod at 76.7 deg,	Pod cheek R)
OPS torso	288 mm	(pod at 95.0 deg,	Pod cheek R)
PILOT neck	296 mm	(pod at 88.1 deg,	Pod cheek L)
PILOT forearm	303 mm	(pod at 95.0 deg,	Pod cheek L)

CRADLE (reclining with the pod parked up) vs fixed structure / pod / dad:

34 mm	Cradle rail L	<-> Nose box rear wall	at 7.5 deg
35 mm	PILOT kid upper arm	<-> Casing front wall L (inner)	at 32.5 deg
35 mm	PILOT kid upper arm	<-> Casing inner wall L	at 35.0 deg
35 mm	Cradle rail R	<-> Nose box rear wall	at 5.0 deg
40 mm	Cradle toe lip	<-> Nose box rear wall	at 0.0 deg
47 mm	PILOT kid foot	<-> Nose box rear wall	at 0.0 deg
60 mm	Kid seat back PILOT	<-> Casing top L (inner)	at 27.5 deg
60 mm	Headrest wing PILOT out	<-> Casing front wall L (inner)	at 30.0 deg
60 mm	Kid seat back OPS	<-> Casing inner wall R	at 35.0 deg
60 mm	Headrest wing OPS out	<-> Casing inner wall R	at 32.5 deg
64 mm	PILOT kid shin	<-> Nose box lid	at 0.0 deg
81 mm	PILOT kid foot	<-> Nose box lid	at 17.5 deg

POD vs structure:

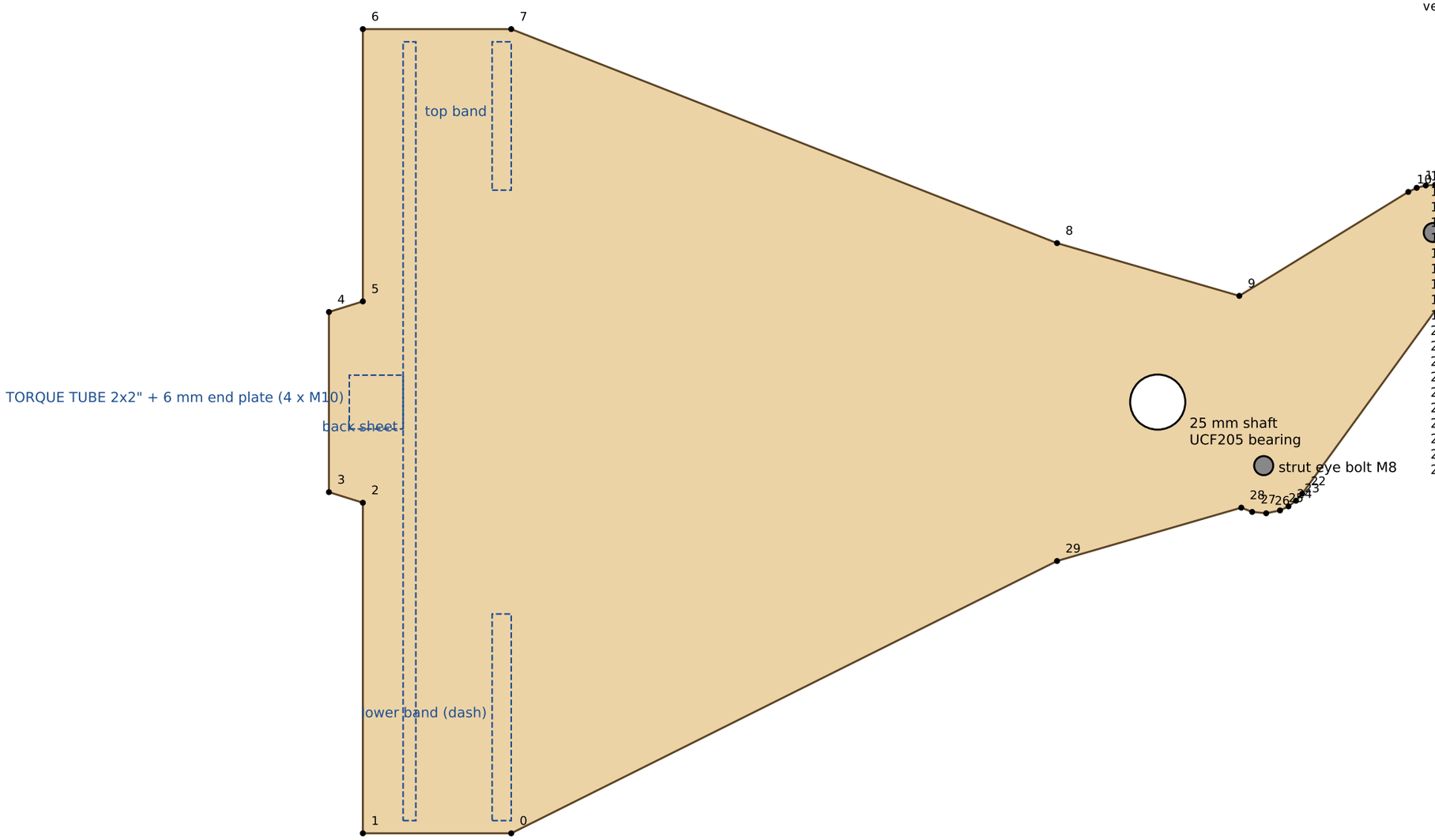
176 mm	Pod cheek L <-> (cradle reclined) Headrest wing PILOT out	at 90.4 deg	
176 mm	Pod cheek L <-> (cradle reclined) Kid seat back PILOT	at 79.0 deg	

Pod cheek in its casing slot: 66 mm to the inner post, 50 mm to the outer post (strut lives there).

HOW EACH PINCH POINT IS HANDLED:

1	pod chin vs knees/shins/hands	: hard stop keeps >= 180 mm (leg); 25 mm foam + NC safety edge on the chin; slow (<= 10 mm/s)
2	pod vs heads	: pivot = head centre -> constant radius, the pod cannot close on a head (>= 250 mm)
3	cheeks in the casing slots	: 48 mm slot, cheek 36 mm -> 6 mm each side (< finger) + brush strip both edges
4	struts / pod actuator	: fully inside the casings (12 mm ply, screwed lids); only the cheek crosses the slot
5	cradle front vs nose box	: >= 30 mm gap all the way (finger 25) + brush strip on the actuator slots
6	cradle sides vs floor	: 6 mm hardboard skirts on the cradle sides overlap the fixed base skins (telescoping guard)
7	under the reclining seat backs	: fixed crush-zone ramp 120 mm below the reclined backs (arm); foam on top
8	seat backs vs casings	: 60 mm side gap, motion is parallel to the wall (no closing edge)
9	kids' arms vs casing front edge	: arm-to-casing >= 35 mm laterally at full recline; 'hands on lap' + dad watches
10	dad	: fixed seat, nothing moves within his reach except the pod back (>= 400 mm from his head)
11	cables across both pivots	: printed pivot clamps <= 60 mm from the axis + 200 mm U service loop in braided sleeve

6. Pod cheek (2 per side, 18 mm ply, laminated) - cutting template coordinates, pivot at (0,0), screen axis to the left



vertex	r (mm, toward screen)	t (mm, up the screen)
0	610	-407
1	750	-407
2	750	-95
3	782	-85
4	782	85
5	750	95
6	750	352
7	610	352
8	95	150
9	-77	100
10	-236	198
11	-244	202
12	-253	204
13	-262	205
14	-275	203
15	-283	199
16	-290	194
17	-298	184
18	-304	171
19	-305	163
20	-304	149
21	-299	137
22	-136	-86
23	-130	-93
24	-123	-98
25	-115	-102
26	-102	-105
27	-89	-104
28	-79	-100
29	95	-150

Latch plate (6 mm steel, bolted to the cheek, centred on the pivot): notches at the UP and DOWN angles.
Stop pin (12 mm steel) through the crank lobe hits rubber-faced steel angles in the casing at both ends.
Print this sheet 1:1 in tiles or transfer the coordinates; both cheeks of a side are glued + screwed together (36 mm).

7. Panel bay detail - how every printed panel bolts to the plywood bands (40 mm grid, M5 into T-nuts)

